

Statement on the use of AI tools

ECO621 Development Economics. Assignment: Putting Numbers to the Nattrass Story. Submitted alongside the main report and the Python scripts.

Tool used

- Claude (Anthropic), run as a coding agent with access to the project working directory, able to read and write files and to run Python.

What it was used for

- Writing and debugging the Python scripts that build the dataset, run the regressions and generate the figures and documents.
- Reading long primary sources and locating figures inside them, in particular the SARB Annual Economic Reports for 2000 and 2001 and the 1996 GEAR document.
- Cross-checking the supplied csv against those primary sources, which is how the inflation-target error described in Part 1 was found.
- Drafting and editing prose, including checking the Part 4 word count and rewriting the report for plainer English.

What it was not used for

- No figure, coefficient, p-value or table entry in this report was produced by the model from memory or estimation. Every number comes from running the included scripts on the supplied data files, or from a named primary source.
- Where a figure could not be sourced, it was left blank rather than filled. Two cases: non-agricultural formal employment growth for 2000, and the 2016 to 2024 employment years.
- The interpretation, the choice of specification, the decision about which vintage to judge GEAR by, and the conclusions are my own.

Verification I carried out

- The 2000 CPI figure and the 2000 nominal Bank rate were each confirmed against two independent sources before use: the SARB Annual Economic Report 2001 narrative, and the monthly Bank rate series supplied in the data folder, which average to 11.81 per cent for 2000 and agree with the report's stated rate changes.
- The corrected GEAR inflation targets were checked directly against the Integrated Scenario table in South Africa (1996).
- All cells flagged as reconstructed from a damaged scan were checked against Nattrass (2001).
- The deficit series was reconciled against its own components: the cash-flow balance divided by nominal GDP reproduces the supplied ratio.

Reproducibility

- All scripts are included with the submission. Running them in the order below regenerates every table, figure and document from the supplied data files.

```
python part1_rebuild_table.py
python part1_table_to_docx.py
python part2_build_extended_series.py
python part2_plot_series.py
python part2_analysis_to_docx.py
python part3_build_regression_data.py
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python part3_estimate.py
python part3_plot_coefficients.py
python part3_analysis_to_docx.py
python part4_to_docx.py
python build_consolidated_report.py
python build_ai_disclosure.py
```

Working directory contents

Listing of the output directory produced for this assignment.

Development project/output/

data/

gear_target_vs_actual_1996_2000_CORRECTED.csv (3,122 bytes)
 part1_gear_vs_reality_table.csv (612 bytes)
 part2_extended_series_1996_2024.csv (26,673 bytes)
 part3_regression_data.csv (8,566 bytes)

figures/

part2_deficit_pct_gdp.png (61,944 bytes)
 part2_inflation_cpi_pct.png (70,674 bytes)
 part2_nonagric_formal_employment_growth_pct.png (51,715 bytes)
 part2_real_bank_rate_pct.png (67,855 bytes)
 part2_real_gdp_growth_pct.png (68,253 bytes)
 part2_real_private_investment_growth_pct.png (71,043 bytes)
 part3_coefficient_stability.png (75,029 bytes)

notebooks/

01_part1_gear_table.ipynb (8,738 bytes)
 02_part2_extend_series.ipynb (14,575 bytes)
 03_part3_regression.ipynb (16,839 bytes)

report/

ECO621_Nattrans_assignment_REPORT.docx (247,518 bytes)
 part1_gear_vs_reality_table.docx (38,154 bytes)
 part2_analysis_draft.md (4,229 bytes)
 part2_extended_series_analysis.docx (412,760 bytes)
 part3_analysis_draft.md (7,980 bytes)
 part3_regression_analysis.docx (108,675 bytes)
 part3_results.txt (7,295 bytes)
 part4_counterfactual.docx (38,939 bytes)
 part4_counterfactual_draft.md (4,109 bytes)
 traps_2_and_3_notes.md (5,338 bytes)

scripts/

build_ai_disclosure.py (6,273 bytes)
 build_consolidated_report.py (31,796 bytes)
 part1_rebuild_table.py (9,400 bytes)
 part1_table_to_docx.py (6,728 bytes)
 part2_analysis_to_docx.py (9,776 bytes)
 part2_build_extended_series.py (9,796 bytes)
 part2_plot_series.py (3,447 bytes)
 part3_analysis_to_docx.py (14,836 bytes)
 part3_build_regression_data.py (5,031 bytes)
 part3_estimate.py (10,097 bytes)
 part3_plot_coefficients.py (3,044 bytes)
 part4_to_docx.py (7,396 bytes)